

APPROVED

GA in BEng (Hons) Engineering: Instrumentation, Measurement and Control
School of Computing, Engineering and Technology

Version	3.0
Course Type	Undergraduate
Award Type	Graduate Apprenticeship in Bachelor of Engineering with Honours - GA in BEng (Hons)
Awards	
No Course Award Assigned	
Code	0631
SCQF Level	Level 10
QAA Subject Benchmark	Engineering (2023)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2026 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100163	electrical and electronic engineering	electrical and electronic engineering	100%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

The four-year Graduate Apprenticeship in BEng (Hons) Engineering: Instrumentation, Measurement and Control is an industry focused degree route, where students work full-time for a partner employer while studying towards a fully accredited degree. The course combines workplace training with a programme of education designed to meet students' career ambitions. The core curriculum is focused on applying the principles of instrumentation, measurement and control coupled with a "flipped-classroom" approach to work-based learning developed through ongoing collaboration between University staff, Workplace Mentors, and line managers in the workplace.

Students will carry out work relating to the instrumentation and control of equipment and processes in the workplace and this can contribute to credit towards the modules. Learning support materials are provided through the VLE which includes a range of equipment such as video and distance learning technologies as well as online interaction. Where appropriate, modules use technologies such as computer-mediated formative assessment to bring interactivity and immediacy to the teaching and learning process. Additionally, there

What the Course Involves	will be opportunities for the student to attend on-campus events to allow face-to-face interaction with staff and peers, participate in group activities with peers and this will contribute to the overall student experience. Arrangements for these on-campus events will be determined in liaison/partnership with employers and stakeholders will be advised well in advance to facilitate attendance. Staff will support students engaged in problem solving from an academic and/or theoretical perspective, while Workplace Mentors will guide students in the application of theory to workplace scenarios. University staff will undertake visits to the student's place of work in order to observe the work-based learning environment and discuss the student's progress with the Workplace Mentor. This combination of blended academic study will allow the student to contextualise learning immediately within their workplace thereby developing their understanding and enhancing and their contribution to organisational effectiveness.
UCAS Code	Code
Course Collaborations	
No collaborations listed	
Introduction to the Course	Graduate Apprenticeships provide work-based learning opportunities for employees and have been created by Skills Development Scotland in partnership with industry and the Further and Higher education sector. Graduate Apprenticeships combine academic knowledge with work-based learning to enable students to become more effective and productive in the workplace. They provide a new way into degree-level study for individuals who are currently employed, and afford the opportunity to undertake degree study whilst working and earning for those individuals who may otherwise have sacrificed Secondary or Further education to enter full-time work. They enable employers to develop their workforce and support staff to build their skills to industry and professional standards. Since Graduate Apprenticeships are designed around the needs of industry, employers can have confidence that what staff are learning will directly contribute to the success of the business. As they are in continuing employment, students can directly apply academic learning to workplace situations, thus truly bringing their studies to life. The BEng (Hons) Engineering: Instrumentation, Measurement and Control covers a broad variety of topics relevant to practice as a professional engineer working in instrumentation, measurement and control, process control engineering and commissioning engineering. This includes theoretical underpinning from foundation mathematics, electronics, and physics, supporting real-world considerations such as actual industry scenarios, international and industrial standards, and serves to develop lifelong learning skills that students will need to stay abreast of the rapidly evolving technologies in instrumentation and control. Through the work-based learning approach offered in Graduate Apprenticeships, and with the School of Computing, Engineering and Technology's close ties to industry, this course is uniquely placed to offer students industry-relevant content and real-world experience with a strong emphasis on control and metering topics. The School already benefits from a wealth of experience in delivering Online Distance Learning (ODL) courses, which will be leveraged in supporting the delivery of course content for the Graduate Apprenticeship. This course has been designed to fully embrace the principles of work-based learning. The course will be delivered over four years with apprentices undertaking 120 credits per year over 12 months. Learning in the workplace will be used to make use of workplace tasks that the apprentice undertakes. The course embraces the principles of work-based learning throughout since all modules will include elements of learning that will be gained directly in the workplace. This encourages the student to reflect on their personal development and they are expected to use an e-portfolio to record their professional and personal development. In addition, it is anticipated that students will undertake the majority of their learning and assessment-related activities in the workplace. Students will draw on their work-based learning experience, and be given opportunities to develop a wide range of knowledge, understanding, and skills. This is made possible through

	the bespoke personal learning plan each apprentice develops in collaboration with their Workplace Mentor and academic supervisor. The learning plan will outline the workplace-specific manner in which the learning outcomes for the modules will be achieved by the apprentice, with the support of their Workplace Mentor.
Educational Aims of the Course	The purpose of the BEng (Hons) Engineering: Instrumentation, Measurement and Control is to provide students with a practical industry relevant qualification integrated with an understanding of general engineering, measurement principles, and classical and modern control theory, instrumentation, industrial automation sensing and measurement systems for monitoring and control and safety applications. This knowledge will be reinforced with the opportunity to develop strong personal, interpersonal, team working and project management skills enabling apprentices to become professional engineers equipped to perform effectively in the engineering workplace. Specifically, graduates of the course will possess the following attributes: - The ability to design, install, commission and maintain control and safety instrument systems including troubleshooting as part of a lifecycle maintenance programme including adherence to national and international standards; - A thorough grounding in the principles, technology and practices of measurement, with an emphasis on the specification, installation, operation and asset lifecycle management of the common types of instrumentation used in industry; - Knowledge of the range and use of analytical methods available for the design, specification, manufacture, and monitoring of instrumentation and control systems; - The ability to apply principles of modelling, classical and modern control concepts and controller design packages in various areas of industry; - An all-round knowledge of instrumentation and measurement systems including remote sensing, and their application, as well as current technological developments and applications to local, regional and global problems; - The ability to design technical flow diagrams with computer-assisted engineering and design software for plant related process control, and instrumentation systems; - A strong theoretical understanding in measurement principles, enabling the apprentice to respond positively to technological development and innovations; - An awareness and appreciation of the practical issues involved in the application of theoretical principles to solving real-world problems within the context of instrumentation, measurement, and control; - The ability to utilise modern, industrially-relevant condition-monitoring and data acquisition systems. The Graduate Apprenticeship student journey differentiates itself from the traditional course of study in that students will be employed for the entire four-year duration of their educational journey. Their employment environment affords them a unique opportunity to take full advantage of work-based learning opportunities, appreciate the relevance of theoretical learning, and open up future career opportunities with their employer. Activities within the employing organisation are coordinated to reflect the content of the modules in which students are enrolled, and are designed to provide the practical application of the theory received through academic learning. The work-based learning within the course aims to offer a new learning experience to meet the needs of both the individuals and employers. This will play a role in achieving the aspiration of the Scottish Government to see a culture of real partnership between employers and education and increasing skill levels in the workforce. This is entirely in keeping with the commitments of the University to 'provide stimulating programmes to anticipate and meet education and skills needs'; to 'strengthen partnerships further with employers and professions in design and delivery models to ensure relevant skills and capabilities' and 'to connect students and industry through enhanced work-related activities'.
Awarding Institution	Robert Gordon University
Course Duration	Work-Based Learning Full-Time - Four years full-time.

Distinctive Features of the Course

Robert Gordon University Graduate Apprenticeships will share institutional Graduate Apprenticeship characteristics; therefore course design will:

- optimise candidate time within the workplace;
- be flexible, enabling students to participate in the course and student experience whilst not bound to a physical campus;
- include meaningful and substantial employer engagement from the outset;
- use innovative teaching, learning and assessment methods;
- support the development of reflective practitioners, equipped to excel in their field and add value to their employers;
- ensure that the professional practice of learners is informed and enhanced by expert technical and theoretical knowledge;
- combine work-based learning, the latest thinking within the field, and work-related/ centric assessments to produce graduates equipped with valuable skills and knowledge.

The Graduate Apprenticeship model of achieving a degree through work-based learning has been designed such that:

- In contrast to the traditional student journey, Graduate Apprenticeship students will be employed for the entire duration of the course, and the course will be designed to integrate their course learning outcomes with suitable workplace projects in collaboration and partnership with their employer. The development of transferable skills will also be supported throughout the course via course content, assessment, and work-based learning.
- The course will utilise the technology investment that the University has made in lecture capture and Virtual Learning Environment (VLE) delivery providing the student with a cutting edge experience.
- Graduate Apprenticeships have been developed specifically to address government desire for programmes which support regional economic development by tackling recognised skill shortages and in furtherance of government youth strategy.
- Despite a number of initiatives to encourage wider access, there are still many young people who would rather go into employment, rather than delaying their earning potential by investing in full-time, higher education. A work-based learning degree programme makes university education available to these people by taking high level learning into the workplace.

Additionally the development of a stream of work-based learning degrees is in line with key strategic aims of the University, particularly to:

- Inspire all individuals to achieve their maximum potential and excel in the world of tomorrow.
- Enable individuals to achieve their ambitions throughout life whatever their circumstances.
- Support students and graduates to enjoy and prosper within a vibrant learning community.
- Contribute to and be an active leader in the development of the regional community.

Employers are closely involved with the specification and delivery of work-based learning activities to support the learning and development of students. Workplace Mentors appointed by the employer, and supported by the University, will work with students to create individual learning plans which outline the shadowing, observation, training, work activities and project opportunities which students will undertake in the workplace. These learning plans will be reviewed by the University to ensure that the required employer contextualisation is appropriate.

The course adopts a practical approach to developing students' professional skills; for example employing group discussion activities within course workshops to foster leadership, listening, and communication skills, and requiring students to deliver presentation and reports as a means of further refining their professional skillset. Within the work-based learning environment, students will continue their professional development as they enhance planning and negotiation skills via development of their learning plan in conjunction with their Workplace Mentor and engage in practical activities designed to further their professional skillset. The knowledge and insights gained across the University and work-based learning environment, coupled with the active role students are expected to take in planning and executing their own learning, will act to supports the students' overall professional enrichment.

	The course is taught by a team of staff who have extensive industrial and commercial experience in instrumentation, engineering principles and measurement and control processes. Activities undertaken by students will be those that have direct relevance to industry, often in the workplace itself.
Placement/Professional Experience	Students enrolled on the course are employed by a collaborating partner organisation while studying for the degree. Employers may provide a series of internal placements within their organisations as part of the student's learning plan where it would be beneficial for the student to spend time within a department other than their own to gain specialist skills and knowledge and where this can be supported by their employer.
Opportunities for Further Study	An Honours degree achieved with at least an upper second classification (2.1) will normally provide access to postgraduate and doctoral programmes in UK higher education institutions.
External and Internal Indicators of Quality and Standards	In October 2016 the QAA (Quality Assurance Agency for Higher Education) issued an Interim Statement on Degree Apprenticeships confirming the final UK-wide statement will be in keeping with the existing suite of characteristics statements within Part A of the UK Quality Code for Higher Education. The QAA further state the Quality Code is a key reference point for setting the academic standards and assuring the quality of apprenticeships involving a higher education qualification. Subsequently, in May 2017 the QAA published its paper 'Quality Assuring Higher Education in Apprenticeships: Current Approaches'. This document highlights considerations for higher education providers in relation to standards and quality and extrapolates relevant guidance from the UK Quality Code. Whilst the document does not constitute formal QAA guidance, its contents have informed the University's approach to the design and delivery of all Graduate Apprenticeships. QAA published engineering Benchmarks (2015) outlining the standards expected for the award of Honours Degrees in Instrumentation, Measurement and Control. The course has been developed in line with the Scottish Credit and Qualifications Framework (SCQF) and its level descriptors, and the Accreditation of Higher Education Programmes (AHEP) 4th edition learning outcomes. This course has been accredited by the Institute of Measurement & Control (InstMC) as meeting the Engineering Council educational requirements for Chartered Engineer (requires further learning).
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards

University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Employment: All applicants must be employed in a full-time position, relevant to their course of study and have the right to live and work in Scotland. An applicant's employer must be committed to and involved in the provision of a suitable workplace environment, coupled with supportive workplace learning guidance and mentoring to create a suitable workplace environment in which the student will be able to obtain the necessary experience and learning outcomes. Employers must be willing to formally partner with the University via a Collaboration Agreement which sets forth the obligations of each party. Subject specific qualifications: SQA Higher: BBCC to include Maths and either Engineering Science, Physics or Technological Studies. English at National 5 grade C or above is required if not held at Higher. Experiential equivalencies: <u>First year entry:</u> For those applicants who do not meet our Standard Course Entry Requirements, applications may be considered from those who possess 1 year of relevant work experience, where the applicant has demonstrated the development of key skills evidenced through a detailed CV. The above are broad standards provided to guide applicants who do not meet our Standard Course Entry Requirements, however in all cases admission decisions rest with the University and the relevance, breadth and depth of employment experience will be taken into consideration. Advanced entry: Applicants who are interested in applying for advanced (or flexible) entry will be considered on an <u>individual</u> basis, applications may be based upon prior education and /or professional experience, with a minimum 3 years relevant work experience and where the applicant has demonstrated the development of key skills evidenced through a detailed CV.. A relevant HNC or HND from one of our partner colleges will be considered for entry into Year 2 or 3. Attention is drawn to the University's process for Recognition of Prior Learning (RPL), and that relevant AHEP4 Learning Outcomes must have been met previously. English language requirements: All applicants whose first language is not English are required to have an Academic IELTS (International English Language Testing System) minimum score of 6.0. Appropriate equivalents will be considered.
Alternative Qualifications for Entry	Alternative qualifications: Applications are welcomed from those with equivalent qualifications to those stated, including Foundation Apprenticeships (SCQF 6), SVQs/NVQs, access programmes and overseas qualifications. Applications from students with non-standard qualifications or work experience will be considered on an individual basis.
Interview Procedures	Interviews may be required on a case-by-case basis.
	In order to progress between stages and achieve an Honours award, students must meet the University AND Engineering Council requirements. Module Pass Criteria In order to pass a module, students must achieve a Grade D. Where a module has more than one component of assessment, students must achieve at least a Grade D in each

Course-Specific Regulations	component. Progression Students must normally complete one stage of study before being allowed to progress to the next stage. Condonement The Assessment Board has discretion to condone a partial failure in a module in accordance with Regulation A4 and providing the Engineering Council requirements, as detailed within their Compensation and Assessment Policy, are met: • The partial failure is at grade E. • All aligned AHEP learning outcomes are met in other successfully completed modules. • The total volume of condoned credit does not exceed 30 credits across all stages, excluding Stage 1. The Assessment Board cannot condone failure of a module delivering AHEP learning outcomes with a mark of more than one grade below the module pass criteria of Grade D. Modules Classed as Central to the Objectives of the Course The requirements of the Engineering Council are that, in order to achieve an accredited Honours award, all major project modules must be passed. The following module has been identified as being central to the objectives of the courses in the programme and, therefore, failure may not be compensated [refer Academic Regulation A4, paragraph 11.1]: EN4103 Honours Project (Graduate Apprenticeship) Awards of Graduate Apprenticeship in BEng (Hons) Engineering The Honours classification is made up of modules from Stage 3 (20% weighting) and Stage 4 (80% weighting) of the course.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is:

University Strategy	• To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
Course-Specific Learning Experience	The introduction of the Graduate Apprenticeship model is a natural development of the University's philosophy to combine theory, practice and experience in developing the skills, knowledge, confidence and professional practice of its students. The principles of work-based learning are core to the philosophy of each Graduate Apprenticeship course at Robert Gordon University. It is fundamental to the Graduate Apprenticeship courses that students have access to relevant and up-to-date theory as they approach practical workplace situations and problems. Encountering theory and practice simultaneously will enable the student to critically analyse each in conjunction with the other. This will embed a rounded knowledge base and develop the ability to question situations while seeking a range of opinions in forming solutions to challenges. Working in conjunction with learning will also enable students to develop the soft skills necessary in an ever-changing work-place. This means that the student will develop not only the knowledge and skills required for the technical aspects of their career but also the ability to operate in the work-place and communicate effectively with internal and external parties. Most of the practical aspects of the course take place in the student's workplace and will therefore vary according to the nature and resources of the student's employing organisation. In addition, the resources of the School of Computing, Engineering and Technology are available to the student and support the teaching staff attached to the course.

The institutional teaching and learning strategy for Graduate Apprenticeships is characterised by a commitment to ensuring a genuine interaction between theoretical and work-based learning. This is supported by a pedagogical approach which embraces and makes use of experiential learning where experiences framed by reflection lead to learning (Schon, 1983) and andragogy where adult learners understand the need to learn something and self-direct their learning process via a negotiated learning plan. Specifically students are encouraged to identify learning opportunities, consider the learning objectives associated with these and reflect upon these experiences to consolidate and apply learning. With regard to work-based learning activities, employers are guided in the identification of suitable activities and projects, and Workplace Mentors are supported in aiding students to reflect on and capture learning from the workplace and encourage the unification of knowledge and experience from the work and university spheres.

The Learning Process: Teaching Strategy and Methods

The course curriculum is designed to deliver the skills and knowledge identified by Skills Development Scotland's framework for Engineering: Instrumentation, Measurement and Control. It has been developed as a coherent whole with an emphasis on the integration of theory, practical skills and applications. It enables the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment. Close consultation with the student's employer will ensure that practical situations and projects encountered within the workplace correlate with the theoretical knowledge developed throughout the course. Opportunities for critical reflection and analysis of apparent or real disparities or conflicts between theory and practice will inform the student of the practicalities of real-world situations and form a rich basis for developing their professional practice.

These skills are developed at levels corresponding to the Scottish Credit and Qualifications Framework levels and form a progression through levels 7, 8, 9 and 10.

As the student progresses through the course they are encouraged to expand their understanding and critical appreciation of instrumentation, measurement and control.

A central feature of all Robert Gordon University Graduate Apprenticeship courses is that they promote a unified, integrated view of the subject area. Courses are designed to:

- Present a coherent underpinning of the theory and practice appropriate to the aims of the course which is further developed and applied within the work-based learning environment. This enables graduates of the course to adapt to future developments in the field.
- Reflect changes within the field and ensure that coverage is given to a selection of emerging topics so that students are aware of likely future developments in the subject together with their potential impact.
- Integrate the delivery of teaching across thematic boundaries to highlight fundamental aspects of interconnectivity and interdisciplinarity within different areas of the syllabus.
- Integrate modes of assessment which complement the methodology of work-based learning and support the development of workplace communication skills.
- Facilitate students to exhibit competence and achievement across a number of different themes (integrated assessment).
- Encourage students to combine elements of their learning from different parts of the course and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment).

The professionally-orientated nature of the course requires students to obtain general and specific skills, this being addressed by a combination of teaching and learning methods. These methods are built around mentored workplace learning directed, supported and extended by University guided, student-centred learning including virtual lectures, tutorials, discussion forums, workshops and private study.

The teaching and learning strategy for GA in Engineering: Instrumentation, Measurement & Control also aligns with that of the School of Computing, Engineering and Technology's teaching and learning strategy which aims to produce industrially relevant professional engineers. In line with the university's teaching, learning, and the student experience strategy, this will enable students to raise their aspirations and thrive in their professional careers as global citizens. RGU continues to lead the way in addressing global sustainability challenges and recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas. Consequently, sustainability is integrated into all courses at the School, equipping students with skills and competencies that are essential for their future careers. The teaching and learning strategy has been designed with reference to engineering council standards to ensure students develop the required professional competencies as they progress through the

Teaching and Learning Strategy

course. This includes a focus on sustainability with students tackling real-world engineering challenges whilst considering sustainable practice. Students are introduced to personal development planning, including professional registration with Engineering Council and relevant Professional Engineering Institutes, and are encouraged to reflect on their own performance and the development of transferrable skills. This puts in place the foundation needed for the lifelong learning and development expected of a professional engineer.

Teaching and Learning Approaches

The main goal of the teaching and learning strategy for the University's Graduate Apprenticeship's is to enable students to become critical, reflective, independent learners who are active participants in the learning process and who develop the motivation and autonomy needed to take control and regulate their own educational experiences and develop the discipline of continuous professional development and life-long learning. This means that, throughout their course of study, students will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their degree, they will be in a position to evaluate and assess their own effectiveness. Students develop the competencies required by a graduate engineer through a wide variety of teaching and assessment methods. These are aligned with the learning outcomes of the modules and the professional competencies students are expected to develop.

This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide-range of appropriate delivery styles. In addition to mentored workplace learning these include conventional instructional teaching methodologies, social learning activities such as group discussion, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning.

Delivery of course materials will be through the University's VLE. Comprehensive course notes and activities will be available for each module. Additionally, recommended reading lists supplement the resources available. Interaction with others is encouraged and achieved primarily through forums and where appropriate, video-conferencing.

Modes of Delivery

Work-Based Learning

The workplace forms the central learning environment for the student. It is important, however, to appreciate that whilst much learning takes place naturally in a workplace that this typically happens in an unplanned and unstructured way, with limited opportunity for reflection or consolidation of learning. For the purposes of this course there needs to be a structure in place to guide the student and ensure that learning outcomes are met. This structure is governed by a subject framework established by Skills Development Scotland and developed into a curriculum by the University who will work closely with the student's employer to ensure that learning outcomes are achieved through correlated theory and practice. This will be expressed for the student in the form of a learning plan which will guide them through the course and help to ensure that workplace learning is supported, resourced, and planned with space and time for critical reflection and consolidation of learning.

Flexible learning materials are available both on and off campus such as textbooks, companion website, videos, formative tests and other online resources. Students will receive a minimum of one visit per year from a member of school staff who will discuss progress with both the student and the Workplace Mentor and will resolve any work related learning issues. In addition, the student will have a direct line of communication with the module teams at all times via email and the University VLE. Module material will be delivered by a mixture of lectures, tutorials and directed study. This will be primarily online and be available to study while the student is in the workplace. Overall guidance will be given on Moodle where students are expected to follow their individualised learning plan supervision from a workplace mentor. When group work is required, teams are expected to develop their own code of conduct that team members will follow and choose their documentation and management standards. Students are expected to develop towards working autonomously and develop a confidence in their ability to work with others and creatively solve problems.

Workplace Mentoring

Key to the delivery and support of work based learning is the Workplace Mentor. Partner employers are required to assign a Workplace Mentor whose role will be to oversee the work based learning undertaken by the student. The University provides employers with guidance as to how appropriate Workplace Mentors should be selected, and provide Workplace Mentors with training and support via induction and ongoing liaison designed to help mentors understand the remit of their role and the activities associated with this. Workplace Mentors will have an overview of the course and its aims, will develop the student's learning plan, assign projects and in-house placements and provide feedback on work-based activities. Workplace Mentors are granted 'Affiliate Student' status enabling them to access all course learning materials via the University's VLE and they may join an online mentor forum. In addition to mentoring support for students, supplementary support may be provided by the employer, for example when subject matter experts are brought in to provide specific business or technical expertise or through access to other departments within the organisation. This enables work-based learning to reflect the range of the curriculum as the student progresses in their studies, therefore providing adequate support for the student's workplace learning on a project, placement or task basis.

Employers' Responsibilities

Partner employers commit to supporting students via the signing of a Collaboration Agreement which sets forth the obligations of the employer and the University. In supporting students, employers are obliged to provide a work based learning environment which adequately represents and reflects the students' course to ensure appropriate functional expertise, supporting infrastructure and professional and technical knowledge within the organisation. Central to this is the provision of a supportive and educative work-based learning environment which affords opportunities to learn, gain new experiences, acquire and build competencies, and reflect on and apply learning. Additionally, employers are obliged to provide protected time to allow students to attend on-campus study days and to undertake study-related activities such as participation in online discussion forums, and access to the requisite IT hardware and software to facilitate this.

Student Role in Learning

Students are expected to be active participants in the learning process. This means that there is a requirement for students to extend their learning from formally delivered materials through wider reading, research and self-directed enquiry. As students develop over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their own personal objectives for learning. Students are encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which they triangulate with feedback from their Workplace Mentor and course delivery team. Students are expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library.

Students are expected to document working practices they have followed. Much of the students independent learning is expected to be spent in team based situations developing elements of the problem solution via collaborative learning. Students are provided with formative and summative feedback via a variety of mechanisms, including during the meeting with their workplace mentors and online discussion forums that enable students to ask questions, exchange knowledge and engage in discussions with their peers and tutors. The students will be encouraged to reflect on theory within their workplace.

Lectures

Lectures are for the purpose of dissemination of information and the demonstration of techniques. These will usually be available via the VLE but may also be delivered face to face at RGU as part of an on-campus event.

Tutorials

Tutorials may be arranged remotely (e.g. online via the VLE) or in person. Tutorials are an opportunity to raise questions, discuss ideas and consequences and exchange experiences and lessons learned. This format is used for a wide range of activities, each suited to the particular subject. It may consist of the staff supporting students engaged in problem-solving or may consist of a series of group exercises where each group is encouraged to allocate responsibilities, allocate tasks, etc. Generally this type of teaching is used to support the lecture,

	clarify the material and experiment with the techniques and skills required. On-Campus Events It is anticipated that from time to time it will be beneficial for Graduate Apprenticeship students to meet on-campus for a variety of events. These may be group sessions where students can discuss the relationship between theory and practice through critical analysis and reflection. On-campus events will also give students the opportunity to develop their professional practice through ideas exchange with peers and academic staff and through other networking. On-campus events will be organised in consultation with employers when there is benefit to be gained for students in attending such events. Individual Project The student will agree a specification for this extended piece of work with their University tutor and their Workplace Mentor. The project allows the student to develop skills in planning, decision making, data collection and analysis by undertaking a major individual engineering research project and reporting the findings of the work. The work will be completed under the guidance and supervision of the student's assigned University tutor.
	The assessment of this course is guided by the course structure and a combination of different types of assessment is used, including project work, coursework, oral assessment, portfolios, online quizzes, and open book tests. The type of the assessment used is determined by the stage of course, SCQF level, learning outcomes and nature of the module content. Assessment will be in line with the University Assessment Policy and quality assurance and will remain the responsibility of the University. Coursework Coursework may be used as a form of summative assessment, and within the context of Graduate Apprenticeship courses this may be work-related, for example based around a project being undertaken in the workplace, or work-centric, such as a report in response to a case study scenario where the case study is built to reflect current issues within a specific sector. Coursework may take many forms, for example reports, essays, lab reports, presentations. Coursework may also include portfolios of evidence demonstrating the application of specific skills and knowledge allowing a view to be formed of the extent to which a student has satisfied learning outcomes. The University provides students with access to ePortfolio software in order to assist students in their creation of portfolios. Where a deadline applies to a particular piece of work the student will be informed at the commencement of the relevant module. Appropriate review measures will be implemented in order to ensure that the students' coursework is individual in nature and not the result of a group effort involving co-workers and colleagues. Work-Based Learning Amongst the assessment tools suitable for the context of work-based learning are diary logs, student blogs, vlogs, other social media engagement, logs developed through critical analysis and reflection which can be worked up as case studies, written critical analysis of theory and practice, reports and presentations. As discussed above, portfolio work also provides an invaluable self-compiled reference resource and starting point for additional learning for the student as they progress into professional practice. Whilst the output of work-based learning may form the basis for assessment, Workplace Mentors will not be expected to play a role in the evaluation of work submitted to Robert Gordon University for summative assessment purposes. Computer Aided Assessment Computer-assisted assessment packages integrated into the VLE may be used for formative and/or summative assessment, to enhance student engagement and increase the speed of

Assessment

feedback.

Unseen Exams

Unseen exams may be used to assess students' base of knowledge and understanding.

Assessed Laboratories

Assessed laboratories may be used to assess students' knowledge, understanding and practical skills, and generally involve students undertaking an unseen practical task under closed book, supervised conditions with a fixed time constraint.

Individual Project

Projects are used in later stages as a learning vehicle and assessment process which allows students to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. Students undertake a major investigative task, using initiative, imagination and creativity; carry out detailed and comprehensive critical analysis of the outcomes.

Assessment Types

Many modules use a combination of assessment types that integrate modes of assessment in order to balance the types of activity and purposes of assessment, such as formative and summative.

Where there are a number of modules that cover allied subject material, integrated assessment may be used to facilitate students to exhibit competence and achievement across a number of different themes.

At all Stages, synoptic assessments embedded into projects and other practical assignments are used to encourage students to combine elements of their learning from different parts of the course and to show their accumulated knowledge and understanding of a set of topics or a subject area.

The Assessment of Student Work

The University has developed a Grading Scheme for assessment purposes, details of which may be accessed at:

<http://www.rgu.ac.uk/academicregulations>

It is important to note that all grades indicated to students during the course of the academic session are provisional until confirmed by an Assessment Board. The Board consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each module undertaken by them and any eligibility for an award. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal.

Any potential barriers to students with disabilities or specific needs are assessed by the School's Disability Coordinator who liaises with the Dyslexia and Disability Service and appropriate teaching staff to ensure that such students are not disadvantaged during an assessment. In the case of examinations, students must register by a specified deadline date to have specific examination requirements considered.

University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	The course will have a dedicated Course Leader to support the development and the delivery of the course. Students will be supported through the School's integration of the institutional Team Around a Student (TAAS) approach, ensuring that every student has a point of contact who can provide advice, triage queries, and connect them with the most appropriate support and resources where issues may be affecting their studies or wider student experience. Module Coordinators are also available to support students with course and module related issues and concerns. Dedicated administrative support will be available to deal with any queries students may have. The School of Computing, Engineering and Technology has a team of Engineering Applications Supervisors and a dedicated flexible learning lab which will be available to students during relevant periods in their studies. The School's e-learning team will also support students and staff as required. The University has a Study Support Service. This provides support in the areas of academic writing, maths and statistics, IT support, English language and Study Skills. In addition to this, students have access to support on effective use of digital learning provided by the Digital Learning Team in Department for Academic Quality and Learning Enhancement (AQLE). Students undertaking individual projects are provided with a detailed Project Handbook covering all aspects of the project operation and assessment. Extensive written guidance is also available via the VLE. All students have an account on the University computer network and can access PCs in the school and across the University. Specialist software is provided on PCs and through virtual machines in Engineering. The support staff also assist students with administrative queries. Frequent informal feedback is provided to students through regular contact with academic staff and Workplace Mentors. Feedback is provided on all assessed work normally within 20

	working days of submission of the work. Provisional grades are made available electronically with final grades available electronically via the Student Portal following ratification by the Assessment Board.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	Annual Course Appraisals are reviewed and approved by the Course Management Team and School Academic Board which meet at least once per semester. The Course Management Team considers teaching and learning issues arising from feedback generated by the External Examiner Annual Reports, Staff and Student Liaison Committees, Student Voice Questionnaires and the National Student Survey. The Course Management Team and School Academic Board also review all proposed changes to the structure and content of courses. The School's Teaching, Learning & Assessment Committee regularly looks at metrics such as student experience and student achievement to ensure that the courses are continuing to deliver the required standard set by the University. It also supports the ongoing enhancement of approaches to teaching, learning and assessment. The quality and standard of the assessment are ensured through a robust moderation process in the School. All summative assessments are internally moderated by an academic staff and Stages 3 and 4 exam papers and assessment briefs are also moderated by an external examiner. The marking of all assessments is completed in full compliance with the University's Assessment policy.
	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements.

University Student Feedback Mechanisms	This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	The Staff Student Liaison Committee meets at least once per semester. The membership of the Committee consists of the Course Leaders, Year Tutors, Student School Officers and student representatives from each Stage of the undergraduate courses. The committee meeting is chaired by the Student School Officer and student representatives are asked to collect comments from their class. However, student representatives are encouraged to liaise with Course Leaders/Year Tutors when issues arise (rather than waiting for the next Staff Student Liaison Committee meeting). Minutes of the meetings are posted on Moodle for all students on the course to access.

Awards

Stage 1

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Higher Apprenticeship in Certificate of Higher Education - Higher Apprenticeship in CertHE	Engineering: Instrumentation, Measurement and Control	Aberdeen, United Kingdom	Exit Award		120

Stage 2

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Higher Apprenticeship in Diploma of Higher Education - Higher Apprenticeship in DipHE	Engineering: Instrumentation, Measurement and Control	Aberdeen, United Kingdom	Exit Award		240

Stage 3

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Graduate Apprenticeship in Bachelor of Engineering - Graduate Apprenticeship in BEng	Engineering: Instrumentation, Measurement and Control	Aberdeen, United Kingdom	Exit Award		360

Stage 4

Permissible Awards	Title	Location	Target Award	PSRB	Award Credits
Graduate Apprenticeship in Bachelor of Engineering with Honours - Graduate Apprenticeship in BEng (Hons)	Engineering: Instrumentation, Measurement and Control	Aberdeen, United Kingdom	Target Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	Apply knowledge of mathematics, statistics, natural science and engineering principles to broadly-defined engineering problems. (F1)	SC1 SC2	SCQF 7, SCQF 7
LLO2	Analyse broadly-defined engineering problems reaching substantiated conclusions. (F2)	SC1 SC4	SCQF 7, SCQF 7
LLO3	Use appropriate computational and analytical techniques to model broadly-defined engineering problems. (F3)	SC2 SC4	SCQF 7, SCQF 7, SCQF 7
LLO4	Apply knowledge of mathematics, statistics, natural science and engineering principles to broadly-defined engineering problems. Some of the knowledge will be informed by current developments in the subject of study. (B1)	SC1 SC2	SCQF 8, SCQF 8
LLO5	Analyse broadly-defined engineering problems reaching substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles. (B2)	SC1 SC3	SCQF 8, SCQF 8
LLO6	Select and apply appropriate computational and analytical techniques to model broadly-defined engineering problems, recognising the limitations of the techniques employed. (B3)	SC2 SC4	SCQF 8, SCQF 8
LLO7	Select and evaluate technical literature and other sources of information to address broadly-defined problems. (B4)	SC1 SC3	SCQF 7, SCQF 7, SCQF 7, SCQF 7, SCQF 8, SCQF 8, SCQF 8
LLO8	Design solutions for broadly-defined problems that meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards. (B5)	SC2 SC5	SCQF 7, SCQF 7, SCQF 8, SCQF 8
LLO9	Apply an integrated or systems approach to the solution of broadly-defined problems. (B6)	SC2	SCQF 7, SCQF 8
LLO10	Evaluate the environmental and societal impact of solutions to broadly-defined problems. (B7)	SC3	SCQF 7, SCQF 7, SCQF 8
LLO11	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct. (B8)	SC3 SC5	SCQF 7, SCQF 7, SCQF 8, SCQF 8
LLO12	Identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity (B9)	SC3	SCQF 7, SCQF 8
LLO13	Adopt a holistic and proportionate approach to the mitigation of security risks (B10)	SC3	SCQF 7, SCQF 8
LLO14	Recognise the responsibilities, benefits and importance of supporting equality, diversity and inclusion. (B11)	SC2 SC5	SCQF 7, SCQF 7, SCQF 7, SCQF 8, SCQF 8
LLO15	Use practical laboratory and workshop skills to investigate broadly-defined engineering problems. (B12)	SC2	SCQF 7, SCQF 7, SCQF 8
LLO16	Select and apply appropriate materials, equipment, engineering technologies and processes. (B13)	SC2	SCQF 7, SCQF 7, SCQF 8
LLO17	Recognise the need for quality management systems and continuous improvement in the context of broadly-defined problems. (B14)	SC1 SC3	SCQF 7, SCQF 7, SCQF 7, SCQF 8, SCQF 8
LLO18	Communicate effectively with technical and non-technical audiences. (B17)	SC4	SCQF 7, SCQF 8
LLO19	Apply knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Some of the knowledge will be at the forefront of the particular subject of study. (C1)	SC1 SC2	SCQF 7, SCQF 9, SCQF 9, SCQF 10, SCQF 10
LLO20	Analyse complex problems to reach substantiated conclusions using first principles of mathematics, statistics, natural science and engineering principles. (C2)	SC1 SC3	SCQF 7, SCQF 9, SCQF 9, SCQF 10, SCQF 10

LLO21	Select and apply appropriate computational and analytical techniques to model complex problems, recognising the limitations of the techniques employed. (C3)	SC2 SC4	SCQF 7, SCQF 9, SCQF 9, SCQF 10, SCQF 10
LLO22	Select and critically evaluate technical literature and other sources of information to address and/or solve complex problems. (C4)	SC1 SC3	SCQF 7, SCQF 9, SCQF 9, SCQF 10, SCQF 10
LLO23	Design solutions for complex problems that meet a combination of societal, user, business and customer needs as appropriate. This will involve consideration of applicable health & safety, diversity, inclusion, cultural, societal, environmental and commercial matters, codes of practice and industry standards. (C5)	SC2 SC5	SCQF 9, SCQF 9, SCQF 10, SCQF 10
LLO24	Apply an integrated or systems approach to the solution of complex problems. (C6)	SC2	SCQF 8, SCQF 9, SCQF 10
LLO25	Use a risk management process to identify, evaluate and mitigate risks (the effects of uncertainty) associated with a particular project or activity. (C9)	SC3	SCQF 8, SCQF 9, SCQF 10
LLO26	Evaluate the environmental and societal impact of solutions to complex problems and minimise adverse impacts. (C7)	SC3	SCQF 8, SCQF 9, SCQF 10
LLO27	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct. (C8)	SC3 SC5	SCQF 8, SCQF 9, SCQF 9, SCQF 10, SCQF 10
LLO28	Adopt a holistic and proportionate approach to the mitigation of security risks. (C10)	SC3	SCQF 8, SCQF 9, SCQF 10
LLO29	Adopt an inclusive approach to engineering practice and recognise the responsibilities, benefits and importance of supporting equality, diversity and inclusion. (C11)	SC2 SC5	SCQF 9, SCQF 9, SCQF 10, SCQF 10
LLO30	Use practical laboratory and workshop skills to investigate complex engineering problems. (C12)	SC2	SCQF 9, SCQF 10
LLO31	Select and apply appropriate materials, equipment, engineering technologies and processes, recognising their limitations. (C13)	SC2	SCQF 9, SCQF 10
LLO32	Discuss the role of quality management systems and continuous improvement in the context of complex problems. (C14)	SC1 SC3	SCQF 8, SCQF 9, SCQF 9, SCQF 10, SCQF 10
LLO33	Apply knowledge of engineering management principles, commercial context, project and change management, and relevant legal matters including intellectual property rights. (C15)	SC1 SC2 SC5	SCQF 8, SCQF 8, SCQF 8, SCQF 9, SCQF 9, SCQF 9, SCQF 10, SCQF 10, SCQF 10
LLO34	Function effectively as an individual, and as a member or leader of a team. (C16)	SC5	SCQF 8, SCQF 8, SCQF 9, SCQF 10
LLO35	Communicate effectively on complex engineering matters with technical and non-technical audiences. (C17)	SC4	SCQF 8, SCQF 9, SCQF 10
LLO36	Plan and record self-learning and development as the foundation for lifelong learning/CPD. (C18)	SC5	SCQF 8, SCQF 8, SCQF 9, SCQF 10

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time Work Based Learning	

Stage 1 / Semester 1**Full Time Work Based Learning**

<i>Core</i>								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	EN1106	Engineering Science		Ioannis Agouris (shsia)	Yes	2.0	30	Level 7
Full Time Work Based Learning	EN1100	Fundamentals of Engineering Practice		Benjamin Bird	Yes	2.0	30	Level 7

Stage 1 / Semester 2**Full Time Work Based Learning**

<i>Core</i>								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	EN1103	Electrical and Electronic Design Principles		Melissa Donald (insmw)	Yes	2.0	30	Level 7
Full Time Work Based Learning	EN1104	Introduction to Industrial Control		Radhakrishna Prabhu (ensrp)	Yes	2.0	30	Level 7

Stage 2 / Semester 1**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CE2107	Embedded Control Design Project		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 8
Full Time Work Based Learning	EN2108	Engineering Mathematics		Anil Prathuru (ap7527)	Yes	2.0	30	Level 8
Full Time Work Based Learning	CE2104	Linear Control Systems		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 8

Stage 2 / Semester 2**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CE2106	Sensor Networks and Data Transfer		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 8

Stage 3 / Semester 1**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	EN3104	Process Control Systems		Adrian Rendon Nava (ar10710)	Yes	2.0	30	Level 9
Full Time Work Based Learning	CE3105	Project and Safety Management		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 9

Stage 3 / Semester 2**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	EN3106	Metrology and Industrial Installations		Somasundar Kannan (sk10513)	Yes	3.0	30	Level 9
Full Time Work Based Learning	EN3103	Workplace-Related Group Project		Nazila Fough (nf8035)	Yes	2.0	30	Level 9

Stage 4 / Semester 1**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	EN4104	Advanced Control Methods		Somasundar Kannan (sk10513)	Yes	3.0	30	Level 10
Full Time Work Based Learning	EN4107	Engineering Management		Hellen Nabayiga (hn12053)	Yes	2.0	30	Level 10
Full Time Work Based Learning	EN4103	Honours Project (Graduate Apprenticeship)	Module is undertaken over 2 Semesters.	Radhakrishna Prabhu (ensrp)	Yes	3.0	30	Level 10

Stage 4 / Semester 2**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	EN4103	Honours Project (Graduate Apprenticeship)		Radhakrishna Prabhu (ensrp)	Yes	3.0	30	Level 10
Full Time Work Based Learning	EN4109	Mechatronics and Automation		Adrian Rendon Nava (ar10710)	Yes	2.0	30	Level 10

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time Work Based Learning	EN1106	Engineering Science		Ioannis Agouris (shsia)	Yes	2.0	30	Level 7
1	Semester 1	Core	Full Time Work Based Learning	EN1100	Fundamentals of Engineering Practice		Benjamin Bird	Yes	2.0	30	Level 7
1	Semester 2	Core	Full Time Work Based Learning	EN1103	Electrical and Electronic Design Principles		Melissa Donald (insmw)	Yes	2.0	30	Level 7
1	Semester 2	Core	Full Time Work Based Learning	EN1104	Introduction to Industrial Control		Radhakrishna Prabhu (ensrp)	Yes	2.0	30	Level 7
2	Semester 1	Core	Full Time Work Based Learning	CE2107	Embedded Control Design Project		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 8
2	Semester 1	Core	Full Time Work Based Learning	EN2108	Engineering Mathematics		Anil Prathuru (ap7527)	Yes	2.0	30	Level 8
2	Semester 1	Core	Full Time Work Based Learning	CE2104	Linear Control Systems		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 8
			Full								

2	Semester 2	Core	Time Work Based Learning	CE2106	Sensor Networks and Data Transfer		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 8
3	Semester 1	Core	Full Time Work Based Learning	EN3104	Process Control Systems		Adrian Rendon Nava (ar10710)	Yes	2.0	30	Level 9
3	Semester 1	Core	Full Time Work Based Learning	CE3105	Project and Safety Management		Judith Abolle-Okoyeagu	Yes	2.0	30	Level 9
3	Semester 2	Core	Full Time Work Based Learning	EN3106	Metrology and Industrial Installations		Somasundar Kannan (sk10513)	Yes	3.0	30	Level 9
3	Semester 2	Core	Full Time Work Based Learning	EN3103	Workplace-Related Group Project		Nazila Fough (nf8035)	Yes	2.0	30	Level 9
4	Semester 1	Core	Full Time Work Based Learning	EN4104	Advanced Control Methods		Somasundar Kannan (sk10513)	Yes	3.0	30	Level 10
4	Semester 1	Core	Full Time Work Based Learning	EN4107	Engineering Management		Hellen Nabayiga (hn12053)	Yes	2.0	30	Level 10
4	Semester 1	Core	Full Time Work Based Learning	EN4103	Honours Project (Graduate Apprenticeship)	Module is undertaken over 2 Semesters.	Radhakrishna Prabhu (ensrp)	Yes	3.0	30	Level 10

4	Semester 2	Core	Full Time Work Based Learning	EN4103	Honours Project (Graduate Apprenticeship)		Radhakrishna Prabhu (ensrp)	Yes	3.0	30	Level 10
4	Semester 2	Core	Full Time Work Based Learning	EN4109	Mechatronics and Automation		Adrian Rendon Nava (ar10710)	Yes	2.0	30	Level 10

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	Work-Based	Full Time	

Sources of Further Information

#	URL and Description
1	https://www.apprenticeships.scot/ Further information about Degree Apprenticeships in Scotland can be found at the Skills Development Scotland Apprenticeships website: https://www.Apprenticeships.scot/ Robert Gordon University Graduate Apprenticeship information made available to employers and prospective students can be found at: http://www.rgu.ac.uk/business-services/graduate-Apprenticeships Further information about the undergraduate programme in the School of Engineering can be found at the School website: http://www.rgu.ac.uk/subjects/engineering Library resources can be accessed via the Library web portal: https://www.rgu.ac.uk/staff-and-current-students/library Advice for student about study and welfare issues is found at the Student Advice and Support portal: https://www.rgu.ac.uk/student-life/student-advice-and-support/ student-advice-and-support Links to the Academic Regulations governing the course can be accessed at: http://www.rgu.ac.uk/academicregulations A number of external sources were consulted in the process of curriculum development for the Graduate Apprenticeship in BEng (Hons) Engineering: Instrumentation and Control: Skills Development Scotland Graduate Apprenticeship subject frameworks are available at: https://www.skillsdevelopmentscotland.co.uk/publicationsstatistics/publications/?page=1&area[]=14=date-desc For reference, the Modern Apprenticeship Program is available at: http://www.sqa.org.uk/sqa/35912.html